🏆 OTC Desk Agent Arena — Run #145

We turned the reasoning-effort knob on three flash models across all six desk workflows. None of them got reliably better. Higher effort made two of them slower and dearer, DeepSeek by nearly nine times the cost. And the idea that every workflow has its own best effort setting did not survive contact with our own trials.

2026-10-02 · run `#145` (effort arms) with the `high` point re-used from runs `#143` / `#144` · 3 models × 6 workflows × 2 trials per arm · plan committed before launch (`2026-09-30-flash-trio-effort-curves-plan.md`) · objective-only scoring (jury off) · output budget unpinned · 42 of 42 arms scored, one with a single trial · cost at list price (see Methods)

The question

Run #143 put three flash models through the desk's six workflows at `high` reasoning effort. This board asks what the effort setting itself buys. Each model got the levels its own route really offers:

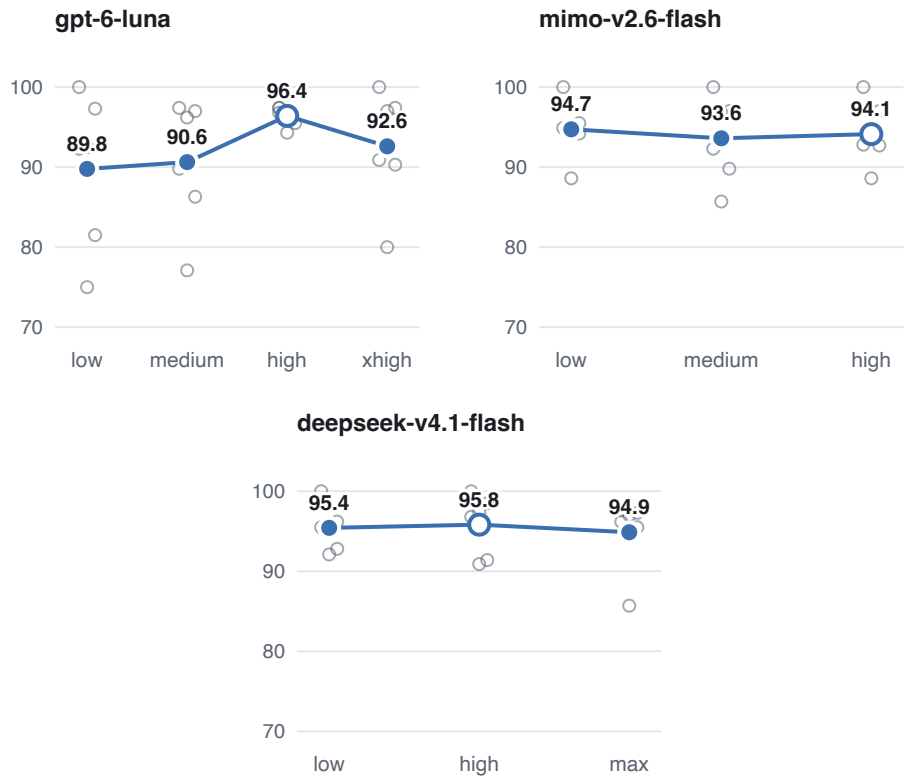
Model	Arms run here	Re-used	Points on its curve
gpt-6-luna	`low`, `medium`, `xhigh`	`high`	4
mimo-v2.6-flash	`low`, `medium`	`high`	3 (its ladder stops at `high`)
deepseek-v4.1-flash	`low`, `max`	`high`	3 (its real ladder)

DeepSeek's API accepts seven effort names, but its own `/models` endpoint declares three: `low`, `high` and `max`. The other four are aliases that run one of those three, so we ran only the real ones. Arms of the same workflow ran back to back, so the comparisons that matter are within a workflow.

Headline

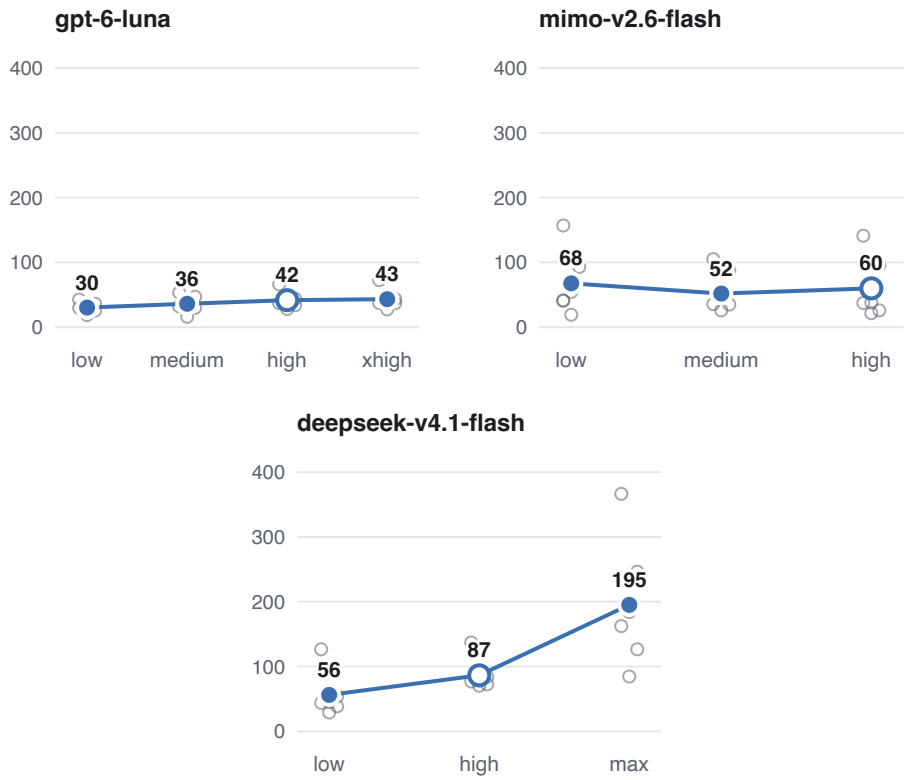
Model · effort	Objective	OVR	EFF	Tool calls	Reasoning tokens	Minutes	List $
gpt–6–luna · low	89.8	79.8	69	30	0.8k	8.4	$0.036
gpt–6–luna · medium	90.6	79.0	48	36	7.2k	10.2	$0.041
gpt–6–luna · *high*	*96.4*	*83.8*	*45*	*42*	*14.9k*	*8.1*	*$0.048*
gpt–6–luna · xhigh	92.6	80.2	34	43	21.8k	14.4	$0.055
mimo–v2.6–flash · low	94.7	82.3	34	68	36.0k	30.4	$0.081
mimo–v2.6–flash · medium	93.6	80.5	40	52	21.2k	22.5	$0.060
mimo–v2.6–flash · *high*	*94.1*	*84.7*	*49*	*60*	*48.7k*	*53.7*	*$0.075*
deepseek–v4.1–flash · low	95.4	**84.3**	40	56	22.3k	7.5	$0.036
deepseek–v4.1–flash · *high*	*95.8*	*79.7*	*1*	*87*	*38.7k*	*9.9*	*$0.049*
deepseek–v4.1–flash · max	94.9	78.0	0	**195**	**509k**	**49.3**	**$0.318**

Italic rows are the re-used `high` point. Objective and OVR are means over six workflow cards; tool calls, reasoning tokens, minutes and cost are per trial.



Objective score by reasoning effort. Line = mean over six workflows; faint circles = each workflow's cell (2 trials). Hollow marker = `high`, re-used from runs #143/#144.

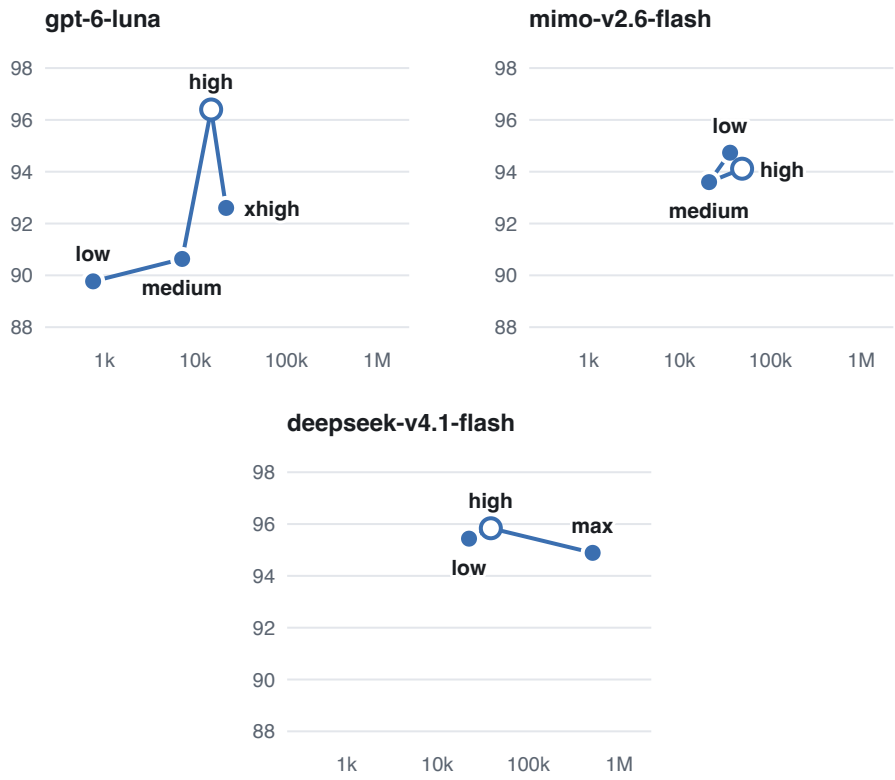
- **DeepSeek: effort is pure cost.** Its objective stays within one point from `low` to `max`. At `max` it reasons 13 times as much as at `high`. Against `low` it makes 3.5 times the tool calls, runs 6.6 times as long and costs 8.8 times as much. `low` has the best card on four of the six workflows and is within a point on the other two.
- **MiMo: flat.** Every arm lands within 1.1 points of the others. `medium` is the cheapest and fastest, `high` has the best card, and none of it is outside noise.
- **Luna: the only curve that moves, and the one we trust least.** Its peak is the re-used `high` point. The section on Luna below explains why we do not read that peak as a property of the setting.



Tool calls per trial by reasoning effort. Line = mean over all trials; faint circles = each workflow's mean. Hollow marker = `high`, re-used from runs #143/#144.

Effort names do not mean the same thing across vendors

`high` on DeepSeek is not `high` on Luna. Measured reasoning tokens are a common unit, so the third figure puts all three models on one log-scale axis.



Objective score against measured reasoning tokens per trial (log scale, shared across panels). Same models, read on a common axis: effort labels are vendor–specific, tokens are not.

Read across, the panels show three different machines. Luna's whole ladder spans under 1k to 22k reasoning tokens per trial. MiMo spends 21k to 49k whatever it is asked. DeepSeek spends 22k to 39k from `low` to `high`, then jumps to 509k at `max`.

That last number corrects something we said before launch. A single-prompt probe found DeepSeek's `max` barely different from `high`. In agentic work it is thirteen times more: the difference compounds over a hundred-odd model calls per trial. **A one-prompt probe tells you whether a setting is accepted. It does not tell you what that setting costs over a workflow.**

Luna's peak is a handful of coin flips

Luna's mean jumps from 90.6 at `medium` to 96.4 at `high` and falls back to 92.6 at `xhigh`. Before reading that as "Luna is best at `high`", we tallied every check across its arms. Of 250 checks, `high` differs from the other arms' average on **15**, and nearly all of them are on two workflows:

- **High-board review (+5.3 net):** whether the model reuses the stored governance report, and whether it writes the board report as an artifact. These steps flip between trials in *every* arm. The artifact step passed at `low` 2/2, `medium` 0/2, `high` 2/2 and `xhigh` 1/2.

- **Trader RFQ (+1.8 net):** the order of the draft and quote calls, and whether it submits for approval. Luna's `low` trials here scored 93.5 and 56.5 on the same inputs.

Luna's trial-to-trial spread is 8.6 points, the largest of the three models. Its `xhigh` arm scored 68.6 and 91.4 on high-board. The `high` arm happens to have won these coin flips twice each, in a different week. We report its curve as **unresolved**, not peaked.

Does each workflow have its own best effort?

A common claim, and the one vendors' guidance leans on, is that the right effort depends on the task. We tested it on this board and on our earlier Luna study, Run #110 (gpt-5.6-luna, `none` / `low` / `high` / `max`, five workflows, two trials).

A table of per-workflow winners always looks task-specific. With two trials per cell, the top cell in each column is partly noise. So we asked two sharper questions.

1. Does the effect of effort differ by workflow more than noise does? This is an effort × workflow interaction. We tested it by permuting the residuals of an additive model (20,000 permutations), so nothing assumes the scores are normally distributed.

2. Would choosing effort per workflow beat one setting for everything? We chose each workflow's best effort from trial A and scored the choice on trial B, then swapped. The alternative chooses one effort for all workflows the same way.

Model	Interaction p (objective / OVR)	Per-task pick vs one setting, on held-out trials (objective / OVR)
deepseek-v4.1-flash	0.35 / 0.22	+1.5 (p 0.05) / 0.0
gpt-6-luna	0.85 / 0.98	−0.9 / −0.1
mimo-v2.6-flash	0.79 / 0.52	+0.7 / +0.8
gpt-5.6-luna (Run #110)	0.23 / 0.15	−0.3 / −3.6

#145 arms only; adding the re-used `high` point changes none of the conclusions (interaction p 0.11–0.96).

- **No interaction is significant**, for any model, on either measure.
- **Choosing per workflow does not pay on held-out trials.** For both Luna studies it is *worse* than one global setting, because the per-workflow winners were mostly noise. Picking with

hindsight on the *same* trial looks much better (gpt-6-luna 96 against 88). That gap is the illusion a per-workflow table creates. DeepSeek's +1.5 is borderline and vanishes once cost counts (OVR 0.0).
- **What this cannot rule out.** Two trials per cell is little power. A simulation at our measured noise shows an 80% chance of detecting a single workflow that gains **17 points** (DeepSeek), 26 (MiMo, gpt-5.6-luna) or **41** (gpt-6-luna) at a higher effort. Smaller task-specific effects may exist. Within this desk's family of workflows, none is large enough to see, or to act on with a desk's realistic budget of trials.

This also corrects our own earlier reading of Run #110. That report said effort's sign is "task-shaped", from per-workflow differences such as high-board gaining 18.6 points at `max`. Under the interaction test (p 0.23) and on held-out trials (−0.3), those differences are within noise.

What the literature says

The claim has real support where tasks differ in **difficulty or kind**:

- Snell et al. (2024) find the compute-optimal test-time budget depends on prompt difficulty.
- Wu et al. (2025) find accuracy rises then falls with chain-of-thought length, with the best length growing with difficulty.
- Gema et al. (2025) find longer reasoning *lowers* accuracy on some task types.
- Anthropic's effort documentation says the impact of effort "varies by task type". OpenAI and Google give similar guidance.

The evidence from **agents** is thinner and mostly recent preprints. It points to the pattern we see:

- Cuadron et al. (2025) find agents that overthink resolve fewer software issues.
- Ares (2026) chooses effort per agent *step* and saves up to 53% of reasoning tokens at little loss.
- Li et al. (2026) find neither more sequential nor more parallel compute reliably helps general agents.

The dominant effect is cost rising while correctness stays flat. DeepSeek's own documentation lists the same collapse of seven effort names into three that we found on its route.

Our reading: the best effort varies between *kinds* of task, such as math against retrieval. Within one family, here six agentic desk workflows, one cheap setting serves every workflow about as well as tuning each one, at a fraction of the cost.

The hypotheses we stated before launch

#	Hypothesis	Verdict
1	Luna repeats Run #110: a gain at the first step up, flat above, `xhigh` costs time	**Partly.** `low` → `medium` is flat (+0.8); `xhigh` takes 71% longer than `low`. The `high` peak is unresolved (see above).
2	DeepSeek's curve is flat, because its effort knob barely moves its reasoning	**Flat: yes. Reason: wrong.** In agentic work `max` reasons 13× as much as `high` and still gains nothing.
3	Effort's sign depends on the workflow	**Not supported** at this power (interaction p ⩾ 0.11 for every model, measure and variant).
4	The direction of tool calls with effort is model–specific	**Yes.** Luna's rise gently (30→43), DeepSeek's more than triple (56→195), MiMo's do not move steadily (68/52/60).

What we would set on the desk

- **deepseek-v4.1-flash: `low`.** Same correctness, the best card, and about a ninth of the cost and a sixth of the time of `max`.
- **mimo-v2.6-flash: `medium` or `low`.** The curve is flat; `medium` is the cheapest and fastest arm on this board.
- **gpt-6-luna: undecided between `medium` and `high`.** Settle it with a same-week board of more trials on high-board and trader RFQ. Those two workflows carry the whole difference.

Methods and caveats

- **Harness defects found during this board, fixed before it finished.** The orchestrator was running two summarizers. The deepagents 0.7 upgrade had appended our compaction middleware instead of replacing the default, so after a compaction the model saw only part of its own history. Separately, a malformed tool call with an id left an unanswered tool result behind. Both broke DeepSeek requests with a 400. Every affected arm (DeepSeek `max` on three workflows, Luna `medium` on one) was set aside with its evidence and re-run

on the fixed code; none of their earlier scores appear here. After the fix, 226 compacted model calls ran with no failed request.

- **The re-used `high` point** ran a week earlier (2026-09-26/27). Since then the only scoring change is the integer-as-string argument fix, already applied to those rows, and every workflow manifest is unchanged. The difference in time is the remaining confound, and it is why Luna's curve is unresolved.
- **One cell has a single trial:** MiMo `medium` on trader RFQ lost its second trial to a network error, excluded by the plan's rules.
- **Cost is list price** at the low end of each route's range. On #143/#144 the billed totals matched it almost to the cent (Luna $0.581, MiMo $0.895, DeepSeek $0.588), so we expect the same here; billed figures will replace these.
- **Two trials per cell** is the board's limit, and the power section above is the honest statement of what that can and cannot show.

Appendix: objective by workflow

Model · effort	Risk-limit breach	Ops settlement	Risk-manager control	High-board review	Trader RFQ	Confirmation desk
gpt-6-luna · low	97.3	100.0	92.3	81.5	75.0	92.5
gpt-6-luna · medium	97.4	89.8	96.2	77.1	86.3	97.0
gpt-6-luna · *high*	*97.4*	*95.5*	*97.4*	*94.3*	*96.8*	*97.0*
gpt-6-luna · xhigh	100.0	90.9	97.4	80.0	90.3	97.0
mimo-v2.6-flash · low	100.0	88.6	94.9	94.2	95.2	95.5
mimo-v2.6-flash · medium	100.0	89.8	92.3	85.7	96.8[1]	97.0
mimo-v2.6-flash · *high*	*100.0*	*88.6*	*93.6*	*92.8*	*92.7*	*97.0*
deepseek-v4.1-flash · low	100.0	92.1	96.2	92.8	96.0	95.5
deepseek-v4.1-flash · *high*	*100.0*	*90.9*	*97.4*	*91.4*	*96.8*	*98.5*
deepseek-v4.1-flash · max	97.3	95.5	96.2	85.7	97.6	97.0

Italic = re-used `high` point. [1] *one trial (network error on the second).*

Rendered from `docs/arena/2026-10-02-run145-effort-curves.md` · OTC Desk Agent Arena · Run #145.